HIMARSHA R. JAYANETTI

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SUMMARY

- **PhD Candidate**: Computer Science, Expected graduation date: May 2027.
- **Publications:** 9 conference papers, 2 journal articles, 5 student conference/poster papers (Google Scholar).
- **Awards:** Best Student Paper (TPDL '22), Best Short Paper (JCDL '23), Best Short Paper Nominee (JCDL '23), and Best Paper Data Science Track and Best Paper Overall (MSVSCC '23 Student Conference)
- **Travel Grants:** Received over $4,000 in competitive travel awards and conference support.
- **Academic Service:** Served on 5 program committees and reviewer roles for conferences including JCDL, TPDL, and CIKM (2021–2026).
- **Internship:** Department of National Archives, Sri Lanka (2016).
- **Industry Experience:** Network Support Engineer at Exetel Telecommunications, Sri Lanka (2017–2019).
- **Internship:** Los Alamos National Laboratory (2022).
- **Internship:** Internet Archive, as a Google Summer of Code Contributor (2025).

RESEARCH INTERESTS

Web Archiving, Digital Libraries, Social Media, Web Science, Data Science, Machine Learning, Information Retrieval, Computational Social Science, Human-Centered AI

EDUCATION

Ph.D., Computer Science, Old Dominion University

Fall 2020 – Present | Norfolk, Virginia, USA

M.S., Computer Science, Old Dominion University

Fall 2019 – Summer 2023 | Norfolk, Virginia, USA

B.Eng., Computer Science, Gujarat Technological University

August 2013 – May 2017 | Ahmadabad, Gujarat, India

WORK EXPERIENCE

Graduate Research / Teaching Assistant, Old Dominion University

Aug 2019 – Present | Norfolk, Virginia, USA

- **Research Assistant:** Conduct research in web archiving, social media analysis, data science, and information retrieval (Spring 2021 – Present, excluding Spring 2026).
- **Teaching Assistant:** CS 120 (Fall 2019; Spring & Fall 2020), **Instructor:** CS 450/550 (Spring 2026).

Google Summer of Code Contributor, Internet Archive

June 2025 – Aug 2025 | Remote

- Developed an LLM-based model for detecting social media content in TV news via logo and post screenshot detection.
- This was a summer internship at the Internet Archive as a Google Summer of Code contributor.

Intern, Los Alamos National Laboratory

June 2022 – Aug 2022 | Remote

- Developed a system to generate social cards for visualizing scientific PDFs.

Network Support Engineer, Exetel Telecommunications

July 2017 – July 2019 | Colombo, Sri Lanka

- Worked with ISP infrastructure and troubleshooting Australia's broadband technologies.

Intern, Department of National Archives

June 2016 – July 2016 | Colombo, Sri Lanka

- Worked on desktop applications, web design, networking, and hardware.

PUBLICATIONS

- Jayanetti, Himarsha R., Sivakanesan Dhanushkanda, Shuai Hao, Michael L. Nelson, and Michele C. Weigle (2026). "Nobody Truly Agrees on Sentiment: Humans, Bespoke Tools, and LLMs Struggle with Social Media Texts". In: *International Conference on Theory and Practice of Digital Libraries*.
- Jayanetti, Himarsha R, Kritika Garg, Sawood Alam, Michael L Nelson, and Michele C Weigle (2024). "Robots still outnumber humans in web archives in 2019, but less than in 2015 and 2012". In: *International Journal on Digital Libraries*, pp. 1–17.
- Jayanetti, Himarsha R (2024). "Evaluating Social Media Reach via Mainstream Media Discourse". In: *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, pp. 5455–5458.
- Jones, Shawn M, Himarsha R Jayanetti, Martin Klein, Michele C Weigle, and Michael L Nelson (2023). "Synthesizing Web Archive Collections into Big Data: Lessons from Mining Data from Web Archives". In: *International Conference on Theory and Practice of Digital Libraries*. Springer, pp. 220–229.
- Jayanetti, Himarsha R, Erika Frydenlund, and Michele C Weigle (2023a). "Exploring Xenophobic Events through GDELT Data Analysis". In: *arXiv*. DOI: 10.48550/arXiv.2305.01708. Modeling, Simulation and Visualization Student Capstone Conference (MSVSCC '23), ODU-VMASC.
- Jayanetti, Himarsha R, Erika Frydenlund, and Michele C Weigle (2023b). "Xenophobic Events vs. Refugee Population–Using GDELT to Identify Countries with Disproportionate Coverage". In: *arXiv*. DOI: 10.48550/arXiv.2308.05038. 16th International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS 2023).
- Garg, Kritika, Himarsha R Jayanetti, Sawood Alam, Michele C Weigle, and Michael L Nelson (2023). "Challenges in replaying archived Twitter pages". In: *International Journal on Digital Libraries* 25.2, pp. 217–236.
- Choudhury, Muntabir Hasan, Lamia Salsabil, Himarsha R Jayanetti, Jian Wu, William A Ingram, and Edward A Fox (2023). "MetaEnhance: Metadata Quality Improvement for Electronic Theses and Dissertations of University Libraries". In: *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. IEEE, pp. 61–65.
- Bragg, Haley, Himarsha R Jayanetti, Michael L Nelson, and Michele C Weigle (2023). "Less than 4% of archived Instagram account pages for the disinformation dozen are replayable". In: *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. IEEE, pp. 102–106.
- Jones, Shawn Morgan, Himarsha R Jayanetti, Alex Osborne, Paul Koerbin, Klein Martin, Michele C Weigle, and Michael L Nelson (2022). "The DSA toolkit shines light into Dark and Stormy Archives". In: *Code {4} Lib* 53.
- Jayanetti, Himarsha R, Shawn M Jones, Martin Klein, Alex Osbourne, Paul Koerbin, Michael L Nelson, and Michele C Weigle (2022). "Creating structure in web archives with collections: different concepts from web archivists". In: *International Conference on Theory and Practice of Digital Libraries*. Springer, pp. 450–458.
- Jayanetti, Himarsha R, Kritika Garg, Sawood Alam, Michael L Nelson, and Michele C Weigle (2022). "Robots still outnumber humans in web archives, but less than before". In: *International Conference on Theory and Practice of Digital Libraries*. Springer, pp. 245–259.

- Garg, Kritika, Himarsha R Jayanetti, Sawood Alam, Michele C Weigle, and Michael L Nelson (2022). "Caching HTTP 404 Responses Eliminates Unnecessary Archival Replay Requests". In: *International Conference on Asian Digital Libraries*. Springer, pp. 329–344.
- Garg, Kritika, Himarsha R Jayanetti, Sawood Alam, Michele C Weigle, and Michael L Nelson (2021). "Replaying Archived Twitter: When your bird is broken, will it bring you down?" In: *2021 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. IEEE, pp. 160–169.
- Choudhury, Muntabir Hasan, Himarsha R Jayanetti, Jian Wu, William A Ingram, and Edward A Fox (2021). "Automatic metadata extraction incorporating visual features from scanned electronic theses and dissertations". In: *2021 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. IEEE, pp. 230–233.

In Review

- Jayanetti, Himarsha R., Sawood Alam, Alexander C. Nwala, Michael L. Nelson, and Michele C. Weigle (2026). "A Multimodal LLM-Based Framework for Detecting Social Media Content in TV News Archives". In: *A Peer-Reviewed Venue*. [Full Paper] (In Review).

ACHIEVEMENTS

ODU Computer Science Graduate Society (CSGS) Hackathon Winner
(2025) Won first place as part of a three-member team (blog).

Best Short Paper Award
(2023) Short Paper at ACM/IEEE Joint Conference on Digital Libraries.

Best Overall Paper and Best Paper in Data Science track Award
(2023) Full Paper at the 16th Annual Modeling, Simulation, and Visualization Student Capstone Conference, Prize: $500 ($350 Best Overall Paper + $150 Best Data Science Track Paper).

Best Student Paper
(2022) Full Paper at Theory and Practice of Digital Libraries Conference, Prize: €500.

Irwin B. Levinstein Scholarship
(2021) Old Dominion University, Funding: $2500.

First Class Honors
(2017) B.Eng., Computer Science, Gujarat Technological University, India.

Nehru Memorial Scholarship
(2013) Indian Council for Cultural Relations and University Grants Commission of Sri Lanka.

TRAVEL GRANTS

NSF DC Travel Award for JCDL
(2026) Funding: $2000 for travel to Dallas, Texas.

CAPWIC Scholarship
(2025) Funding covering conference registration ($50) for Washington, DC.

NSF Travel Grant for CIKM
(2024) Funding: $1250 for travel to Boise, Idaho.

SBP-BRIMS Conference Scholarship
(2023) Funding: $1000 for travel to Pittsburgh, Pennsylvania.

ACM SIGIR Student Travel Grant for JCDL
(2023) Funding covering conference registration and $1420 travel support for Santa Fe, New Mexico.

ODU SEES Travel Award for TPDL
(2022) Funding: $500 for travel to Padua, Italy.

GRANT PROPOSALS

- As a Graduate Research Assistant, I contributed to a grant proposal extending my dissertation research on how social media content extends beyond its native platform and is integrated into television news. This proposal resulted in $750,000 in federal funding from the Institute of Museum and Library Services (IMLS).

TEACHING EXPERIENCE

- **Instructor of Record**, CS 450/550: Database Concepts (Spring 2026)
- **Teaching Assistant**, CS 120: Introduction to Information Literacy and Research (Fall 2019; Spring & Fall 2020)

SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, R, Bash / shell scripting
Frameworks & Tools: Django, NodeJS, React, Docker, Elasticsearch, Kibana, GitHub
Data Science: TensorFlow, WEKA, NumPy, Pandas, NetworkX
Visualization: D3, Vega-lite, Tableau, ObservableHQ
Databases: MongoDB, MySQL, PostgreSQL, SQLite
Systems: Linux / Unix, Cloud platforms (AWS, GCP, Azure), Networking fundamentals (TCP/IP, HTTP, DNS)
Version Control: Git
Machine Learning & AI: NLP, Computer vision, Model evaluation, LLM prompting, Multimodal models
Soft Skills: Technical writing, Research communication, Cross-disciplinary collaboration, Grant writing, Project leadership, Peer code review & mentorship

SERVICES & OUTREACH

- **Program Committee Member**, CIKM 2026 (Full Research, Short Paper)
- **Program Committee Member**, JCDL 2026 (Full or Short Research Papers)
- **Program Committee Member**, TPDL 2026 (Full Papers, Short and Late Breaking Results Papers)
- **Program Committee Member**, JCDL 2025 (Research Paper)
- **Secretary**, Sri Lankan Student Association at ODU (2023-2024)
- **Student Mentor**, NSF Research Experiences for Undergraduates (REU) program (2022)
- **President**, ACM-W at ODU (2022)
- **Secretary**, ACM-W at ODU (2021)
- **Peer Reviewer**, TPDL (2021)
- **Instructor**, Data Science Summer Camp (2021)
- **Volunteer**, STRS Summer Camp (2020)

PROJECTS

Ph.D. Dissertation

2023 – Present

This study examines how social media content extends beyond its platform and is integrated into television news. Existing research largely measures impact through intra-platform engagement, overlooking television's role as a widely consumed and more trusted medium with a distinct audience. As a result, additional reach and influence of social media content beyond its native platform remain significantly underestimated. To address this gap, this work proposes a framework for detecting and analyzing social media references in TV news broadcasts.

What's Missing? Innovating Interdisciplinary Methods for Hard-to-Reach Environments

2022 – 2025

This was a large interdisciplinary research project which was funded by the U.S. Department of Defense Minerva Research Initiative investigating methods for studying hard-to-reach environments (Khayelitsha Site-C, South Africa and Villa Caracas, Colombia). The project brought together an international team across the U.S., Canada, Norway, and Colombia. The work examines methodological and epistemological gaps when only certain research approaches are feasible. Our component focuses on analyzing these sites using public data sources such as global news and social media, alongside complementary methods including surveys, visual sociology, and citizen science across collaborating teams. The project was later discontinued before completion.

Data Science for Social Good Project

2022 – 2023

This project is titled "Data Science for Social Good: Mining and Visualizing Worldwide News to Monitor Xenophobic Violence" and was funded by the 2022-2023 ODU Data Science Seed Funding Program. We examined xenophobic events related to refugees and migration using the GDELT 2.0 database and APIs. Our research used visualizations to explore patterns in media coverage through two case studies. We also discussed the data analysis process and the challenges of working with GDELT data and its tools.

Analyzing Unnecessary Traffic In Web Archives

2022

Archived web pages can generate repeated, invisible HTTP requests, creating unnecessary traffic and potentially overwhelming web archive servers. Pages requiring frequent updates, such as sports scores or playlists, were especially likely to cause this problem, particularly when missing resources produce repeated 404 errors. We proposed to use Cache-Control headers to cache 404 responses, preventing unnecessary requests from reaching the archive server. Our results showed that this approach can reduce wasted network and computational resources during archival replay.

Access Patterns of Robots and Humans in Web Archives

2022

We analyzed human and robot access patterns in web archives using Internet Archive's Wayback Machine and Portuguese Web Archive access logs from 2012 and 2019. We classified user sessions as human or robot based on browsing behavior and then analyzed them to identify navigation patterns and temporal preferences. The study found that robots dominated archive traffic, accounting for up to 98% of requests, while their access patterns became more varied over time. Both humans and robots showed a strong preference for recently archived web pages.

ETD Data Enhancement

2022

We developed MetaEnhance, an AI-based framework for automatically detecting, correcting, and standardizing metadata errors in Electronic Theses and Dissertations (ETDs). Using a benchmark of 500 ETDs, MetaEnhance achieved near-perfect F1-scores for error detection and correctness F1-scores ranging from 0.85 to 1.00 for five of seven fields.

Adverse Effects of Twitter's UI change on Web Archives

2021

We examined the challenges of preserving Twitter content in web archives following Twitter's June 2020 UI change. We found that many archives could not properly capture the new interface, leading to missing or incomplete content and potentially inaccurate historical records. Using the personal Twitter account of the 45th U.S. President, @realDonaldTrump, we identified missing evidence of misinformation labels and temporal inconsistencies in archived pages. The goal of this study is to caution the researchers to critically evaluate archived social media data before drawing historical conclusions.

DSA Project

2021

This project developed storytelling solutions to improve the discovery and understanding of web archive collections. I contributed to several components of the DSA project, including AIU, Raintale, and MementoEmbed. AIU is a Python library that gathers seed URLs and metadata from web archive collections such as Archive-It, Pandora, and Trove using APIs and screen scraping. MementoEmbed generates archive-aware cards for individual mementos, while Raintale organizes these cards into engaging, shareable stories. Taken together, these tools help transform large and complex web archive collections into accessible and meaningful experiences for archivists, researchers, and the public.

Extracting Metadata from Scanned ETDs

2020

We developed a conditional random field (CRF) model that combines text-based and visual features to automatically extract metadata from scanned Electronic Theses and Dissertations (ETDs). We evaluated our method using 500 human-validated ETD cover pages. The model significantly outperformed text-only and heuristic approaches, achieving 81.3%–96% F1 scores across seven metadata fields.

M.S. Thesis

2019 – 2023

My master's thesis titled, "Supporting Account-based Queries for Archived Instagram Posts". We addressed a fundamental challenge in social media preservation: valuable Instagram content exists in web archives, but users often cannot discover it because they do not know the original post URLs. I developed two approaches to enable users to retrieve archived Instagram posts associated with a specific account, one leveraging WARC revisit records within the Internet Archive and the other using an external index mapping Instagram accounts to their posts. I implemented and evaluated both approaches, highlighting their practical advantages and limitations for web archives.

ARTICLES

- **Teaching Database Concepts for Senior Undergraduate and Graduate Students at ODU**
  https://ws-dl.blogspot.com/2026/06/2026-06-01-teaching-database-concepts.html
- **Summer Project as a Google Summer of Code (GSoC) Contributor**
  https://ws-dl.blogspot.com/2025/09/2025-09-29-summer-project-as-google.html
- **A Battle of Opinions: Tools vs. Humans (and Humans vs. Humans) in Sentiment Analysis**
  https://ws-dl.blogspot.com/2025/03/2025-03-26-battle-of-opinions-tools-vs_61.html
- **Tracking Political Trends Around US Presidential Election**
  https://ws-dl.blogspot.com/2025/02/2025-02-11-tracking-political-trends.html
- **Tools for Using the Internet Archive's TV News Archive as Research Data**
  https://ws-dl.blogspot.com/2024/08/2024-08-07-tools-for-using-internet.html
- **Lil Tay's Fake Death Announcement: The Quest for a Deleted Instagram Post in Web Archives**
  https://ws-dl.blogspot.com/2023/08/2023-08-22-quest-for-deleted-instagram.html
- **PDFServer - Our Summer Internship at LANL**
  https://ws-dl.blogspot.com/2022/12/2022-12-21-pdfserver-our-summer.html
- **Digging Up a Gem Through the Web Archives**
  https://ws-dl.blogspot.com/2021/09/2021-09-20-digging-up-gem-through-web.html
- **Creation Time and Published Time Are Not the Same: Estimating the Instagram Epoch**
  https://ws-dl.blogspot.com/2021/02/2021-02-20-creation-time-and-published.html
- **Twitter Rewrites Your URLs, but Assumes You'll Never Rewrite Theirs: More Problems Replaying Archived Twitter**
  https://ws-dl.blogspot.com/2021/01/2020-01-22-twitter-rewrites-your-urls.html
- **Twitter Added Labels On Its Old User Interface**
  https://ws-dl.blogspot.com/2020/12/2020-12-08-twitter-added-labels-on-its.html
- **How Well Is Instagram Archived?**
  https://ws-dl.blogspot.com/2020/11/2020-11-04-how-well-is-instagram.html
- **New Twitter UI: Replaying Archived Twitter Pages That Never Existed**
  https://ws-dl.blogspot.com/2020/11/2020-11-04-new-twitter-ui-replaying.html
- **Twitter Was Already Difficult To Archive, Now It's Worse!**
  https://ws-dl.blogspot.com/2020/07/2020-07-15-twitter-was-already.html

A complete list of articles available at himarshaj.github.io/blogposts